BADMINTON SWING ANALYZER USING MPU6050 WITH ML

MINI-PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled "**BADMINTON SWING ANALYZER USING MPU6050 WITH ML** "is the bonafide work of **Mr.R.ASWIN RAAJ (REG.No:95172023), Mr.S.MANI BALAJI (REG.NO:95172023) and Mr.V.S.THARUN (REG.NO:95172023)** who carried out the mini-project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Examiner 1 **Examiner 2**

ABSTRACT

The rapid advancement of embedded systems and machine learning has enabled the development of intelligent sports analytics solutions. This project presents a Smart Badminton Performance Analytics System designed to monitor, classify, and analyze player performance using real-time motion data. The system utilizes an ESP32 microcontroller integrated with an MPU6050 inertial measurement unit (IMU) to capture acceleration and angular velocity during badminton swings. A threshold-based algorithm is implemented to detect valid swings, and key parameters such as speed, impact, and duration are extracted. These parameters are further processed using feature engineering techniques to derive meaningful metrics such as power, efficiency, and energy. Machine learning models are employed to classify swings into WEAK, MEDIUM, and STRONG categories and to identify stroke types including SMASH, DROP, CLEAR, and DRIVE. A hybrid approach combining machine learning and rule-based logic enhances classification accuracy and system reliability. The processed data is analyzed to generate advanced insights such as player performance level, consistency score, fatigue detection, and comparison with professional benchmarks.The system provides feedback through a web-based dashboard and a Streamlit interface, enabling intuitive visualization of performance metrics. Furthermore, the project outlines a pathway for transitioning from a prototype to a compact wearable product by incorporating ultra-low power microcontrollers such as Seeed Studio XIAO nRF52840 or ESP32-C3, advanced IMU sensors like ICM42688, and lightweight Li-Po battery systems.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Badminton is a fast-paced sport that requires a combination of speed, precision, timing, and power. Performance analysis in badminton is traditionally done through manual observation by coaches, which is often subjective and lacks quantitative accuracy.

With the advancement of embedded systems, sensors, and machine learning, it is now possible to develop intelligent systems that can capture, analyze, and interpret player performance in real time. This project presents a Smart Badminton Performance Analytics System, which integrates sensor data acquisition, embedded processing, and machine learning techniques to evaluate player swings.

The system uses an inertial measurement unit (IMU) sensor to capture motion data during gameplay. The collected data is processed to extract meaningful features such as swing speed, impact force, and duration. These parameters are further analyzed using machine learning models to classify swing performance and provide insights into player technique.

1.2 Problem Statement

Traditional badminton training methods rely heavily on visual feedback and manual coaching, which lack precision and consistency. Players often do not have access to real-time performance metrics such as swing strength, impact quality, and consistency.

There is a need for a system that can:

- Automatically capture swing data
- Provide real-time performance analysis
- Classify player performance accurately
- Offer data-driven feedback for improvement

1.3 Objectives

The main objectives of this project are:

- To design and develop a sensor-based system for capturing badminton swing data
- To implement real-time swing detection using embedded systems
- To extract meaningful features such as speed, impact, and duration

- To develop machine learning models for swing classification
- To analyze player performance using advanced analytics techniques
- To provide feedback and suggestions for performance improvement

1.4 Scope of the Project

This project focuses on developing a portable and cost-effective system for analyzing badminton player performance. The system is designed to work in real-time and provide insights into swing characteristics.

The scope includes:

- Real-time data acquisition using sensors
- Swing classification into categories (Weak, Medium, Strong)
- Stroke type classification (Smash, Drop, Clear, Drive)
- Performance analysis including consistency, fatigue, and stability
- Visualization of results through a web interface and dashboard

The system can be extended to other sports and enhanced with cloud-based analytics and mobile applications.

1.5 Motivation

The motivation behind this project is to bridge the gap between traditional coaching methods and modern data-driven sports analytics. By leveraging embedded systems and machine learning, the project aims to provide players with objective performance metrics and actionable insights.

This system enables:

- Improved training efficiency
- Better understanding of player strengths and weaknesses
- Data-driven decision making in sports training

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The development of intelligent sports analytics systems has gained significant attention in recent years due to advancements in embedded systems, wearable sensors, and machine learning. In badminton, performance analysis traditionally relies on manual observation, which lacks precision and real-time feedback. Researchers and developers have explored various approaches to overcome these limitations using technology.

This chapter reviews existing methods and systems related to motion tracking, wearable devices, and machine learning-based sports analytics.

2.2 Wearable Sensor-Based Systems

In badminton and similar sports, inertial measurement units (IMUs) are used to capture swing motion. These sensors provide raw data such as acceleration and angular velocity, which can be processed to extract meaningful performance metrics.

Limitations:

- Limited analysis capabilities in basic systems. Lack of advanced feature extraction. Minimal or no real-time feedback

2.3 Vision-Based Motion Analysis Systems

Camera-based systems use computer vision techniques to analyze player movements. These systems track body posture, racket movement, and shuttle trajectory using image processing algorithms.

Drawbacks:

- High computational requirements. Dependency on lighting and environment. Not portable

2.4 Machine Learning in Sports Analytics

Machine learning has been widely applied in sports analytics to classify player actions and predict performance. Algorithms such as decision trees, support vector machines, and neural networks are used to analyze sensor data.

In motion-based sports:

- Classification models identify movement patterns. Feature engineering improves prediction accuracy. Data-driven insights help in performance optimization. Focus only on classification. Do not integrate real-time embedded systems. Lack comprehensive analytics (fatigue, consistency, etc.)

2.5 Existing Badminton Analysis Systems

Some existing badminton analysis systems include. Smart rackets with built-in sensors. Mobile applications for swing tracking. Basic motion analysis tools hese systems provide limited features such as, Swing speed measurement. Basic shot detection

Limitations are High cost. Limited analytics capabilities. No personalized coaching insights. Lack of integrated AI-based feedback

2.6 Identified Research Gap

From the study of existing systems, the following gaps are identified:

- Lack of integrated systems combining embedded hardware and machine learning. Limited real-time analytics capabilities. Absence of advanced performance metrics such as:
    - Consistency
    - Stability
    - Fatigue detection
- No comprehensive player performance evaluation
- Lack of intelligent feedback or coaching suggestions

2.7 Proposed System Contribution

To address these limitations, the proposed system introduces:

- A sensor-based embedded system for real-time data acquisition
- A machine learning-based classification system for swing and stroke analysis
- A hybrid approach combining rule-based and ML models
- Advanced analytics including:
    - Player vs professional comparison
    - Consistency and stability analysis
    - Fatigue detection
- A user-friendly interface for real-time monitoring and data visualization

This approach provides a cost-effective, portable, and intelligent solution for badminton performance analysis.

CHAPTER 3
SYSTEM OVERVIEW

3.1 Introduction

The Smart Badminton Performance Analytics System is designed as an integrated platform that combines embedded hardware, data processing, machine learning, and user interface modules to analyze player performance in real time. The system captures motion data from the player's racket, processes it to extract meaningful features, and applies intelligent algorithms to evaluate performance.

This chapter presents the overall architecture and working pipeline of the proposed system.

3.2 Overall System Architecture

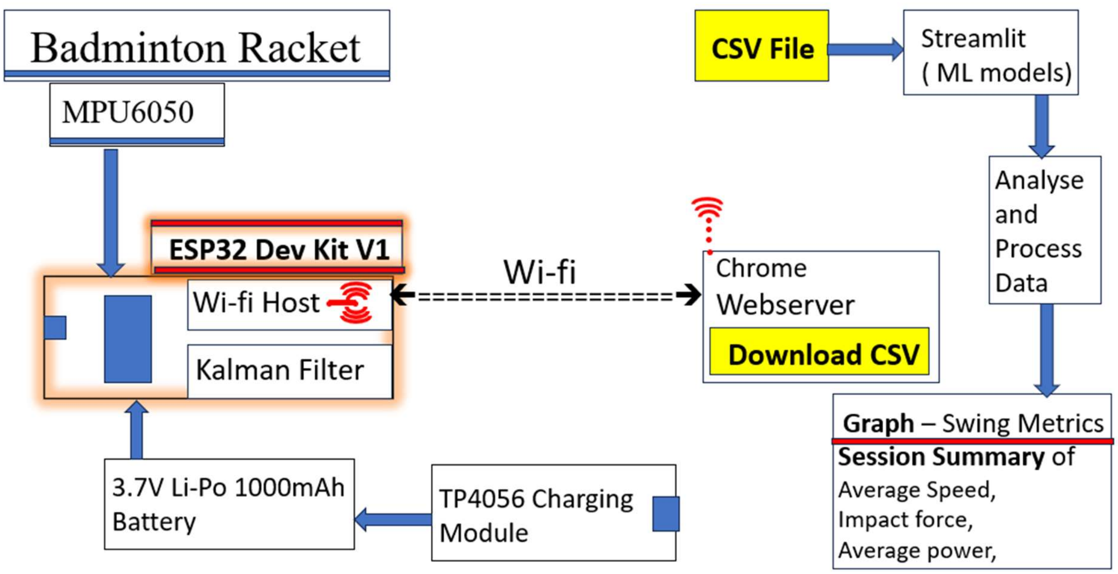

Fig: 3.1 Block diagram of Badminton swing analyzer using mpu6050 with ML

The system consists of multiple interconnected layers that work together to perform data acquisition, processing, analysis, and visualization.

- Sensor Layer (MPU6050)
- Embedded Processing Layer (ESP32)
- Data Processing & Feature Engineering Layer
- Machine Learning Layer
- Analytics & Decision Layer
- User Interface Layer

3.3 Working Pipeline

The complete working of the system is divided into the following stages:

3.3.1 Data Acquisition

- Motion data is captured using the MPU6050 sensor
- Accelerometer and gyroscope provide raw values:
  - Ax, Ay, Az, Gx, Gy, Gz
- ESP32 reads sensor data continuously using I2C communication

3.3.2 Signal Processing

- Raw data is calibrated to remove sensor offsets
- Gravity component is filtered using a smoothing algorithm
- Total acceleration and angular velocity are calculated

These processed values form the basis for further analysis.

3.3.3 Swing Detection

The system identifies a valid swing using threshold-based logic:

- Start Condition: Acceleration exceeds a predefined threshold
- Tracking Phase: Peak values of speed and impact are recorded
- End Condition: Acceleration drops below threshold and minimum duration is satisfied

For each detected swing, the following parameters are extracted:

- Peak speed
- Maximum impact. Swing duration

3.3.4 Data Logging

- Each swing is stored in CSV format
- Data includes:
  - Sensor readings
  - Computed features
  - Timestamp
- Data can be downloaded via the web interface for further analysis

3.3.5 Feature Engineering

The collected data is processed to generate meaningful features:

- Power = Speed × Impact
- Efficiency = Impact / Duration

These features improve the performance of machine learning models and enable deeper analysis.

3.3.6 Machine Learning Processing

The system uses two machine learning models:

- Swing Classification Model
  - Classifies swings into: Weak, Medium, Strong
- Stroke Classification Model
  - Identifies stroke type: Smash, Drop, Clear, Drive

Additionally, a rule-based threshold system is used alongside ML predictions to improve reliability.

3.3.7 Analytics and Decision Making

An advanced analytics engine processes the classified data to generate insights:

- Player vs professional comparison
- Gap analysis
- Consistency and stability evaluation
- Fatigue detection
- Player type classification

Based on these analyses, the system generates performance feedback and improvement suggestions.

3.3.8 User Interface

The system provides multiple interfaces:

- ESP32 Web Dashboard: Displays real-time speed, impact, and swing count
- Terminal Dashboard: Shows detailed analytics results
- Streamlit Application: Provides interactive visualization and insights

3.4 System Modes of Operation

The system operates in different modes:

- Calibration Mode: Used to collect initial data and set thresholds
- Game Mode: Used for real-time performance tracking
- Idle Mode: System remains inactive

3.5 Advantages of the Proposed System

- Integration of embedded systems and machine learning, combining hardware sensing with intelligent analysis
- Hybrid approach (machine learning + rule-based logic) for improved accuracy and reliability
- Advanced analytics beyond basic classification, including consistency, stability, and fatigue analysis
- Custom calibration mechanism for personalized threshold setting based on player data
- Coaching and feedback system for performance improvement

CHAPTER 4

HARDWARE COMPONENTS

4.1 Introduction

The hardware layer forms the foundation of the proposed system. It is responsible for capturing real-time motion data and enabling interaction with the user. The system primarily consists of a microcontroller, motion sensor, input controls, and output indicators.

This chapter describes the components used, their working principles, and the overall hardware setup.

Components

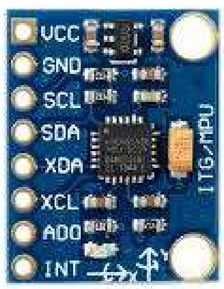

Fig 4.1 MPU6050

Components

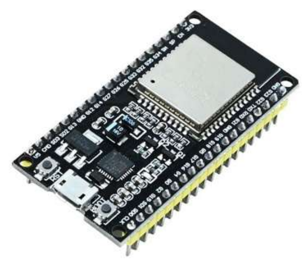

Fig 4.2 ESP32 Dev Kit V1

Fig 4.3 Push Button

Fig 4.4. Li-po 3.7v 1000mah battery

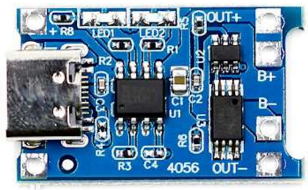

Fig 4.5 TP4056 Charging Module

Fig 4.6 Badminton Racket.

4.2 MPU6050 Sensor Module

With reference to Fig 4.1, the MPU6050 is a 6-axis Inertial Measurement Unit (IMU) consisting of a 3-axis accelerometer and a 3-axis gyroscope. It is the primary sensing element of the system and plays a crucial role in capturing real-time motion data during badminton gameplay.

The accelerometer measures linear acceleration along the X, Y, and Z axes, while the gyroscope measures angular velocity, which represents rotational movement of the racket. These raw sensor readings are continuously transmitted to the ESP32 microcontroller for processing.

In this project, the MPU6050 is responsible for detecting swing dynamics such as speed, impact force, and motion direction. These parameters form the foundation for feature extraction and machine learning classification. Without this sensor, real-time motion analysis would not be possible.

Thus, the MPU6050 acts as the core data acquisition unit of the system.

4.3 ESP32 Dev Kit V1

With reference to Fig 4.2, the ESP32 Dev Kit V1 is the central processing unit of the system. It acts as the brain of the entire setup and is responsible for collecting data from the MPU6050 sensor and executing all processing algorithms.

The ESP32 performs multiple critical functions such as:

- Reading real-time sensor data using I2C communication
- Executing swing detection algorithms
- Performing feature calculations like speed, impact, and power

- Managing mode control (calibration and game mode)
- Hosting a web server for real-time dashboard visualization

Additionally, it enables wireless communication using Wi-Fi, allowing the processed data to be displayed on a web interface. Therefore, the ESP32 serves as the core controller and communication module of the system.

4.4 Push Button

With reference to Fig 4.3, the push buttons are used as manual input interfaces for controlling the system modes. Two buttons are implemented in this project: one for calibration mode and another for game mode.

These buttons provide a simple and effective way for the user to interact with the system without requiring external software control. Hence, they act as the mode selection and control interface of the system.

4.5 Li-Po 3.7V 1000mAh Battery

With reference to Fig 4.4, the Li-Po 3.7V 1000mAh battery is used as the primary power source for the entire system. It provides stable and portable energy required to operate the ESP32 and sensor modules during badminton sessions.

4.6 TP4056 Charging Module

With reference to Fig 4.5, the TP4056 charging module is used for safe and efficient charging of the Li-Po battery. It provides regulated charging with overcharge, over-discharge, and short-circuit protection.

The module allows USB-based charging, making it convenient for users to recharge the system. It ensures battery safety and extends battery life, which is crucial for wearable embedded systems.

Hence, the TP4056 acts as the battery management and protection module.

4.7 Badminton Racket

With reference to Fig 4.6, the badminton racket serves as the physical platform for the entire system. The MPU6050 sensor and embedded components are mounted on the racket to capture real-time motion during gameplay.

The racket is not only a sports equipment but also the data generation platform of the system. Every swing performed by the player is detected through sensor movement attached to the racket, enabling accurate performance analysis.

Thus, the racket acts as the application environment and sensing medium of the system.

CHAPTER 5
SOFTWARE ARCHITECTURE

5.1 Introduction

The software architecture of the Smart Badminton Performance Analytics System integrates embedded firmware, data processing, machine learning models, and user interface modules. The system is designed as a multi-layer architecture where each layer performs a specific function and communicates with other layers to achieve real-time performance analysis.

5.2 Overall Software Architecture

The system is divided into the following layers:

- Embedded Firmware Layer (ESP32)
- Data Processing Layer (Python)
- Machine Learning Layer
- Analytics & Decision Layer
- User Interface Layer (Web + Streamlit)

ESP32 → CSV Data → Python Processing → ML Models → Analytics → Dashboard

5.3 Embedded Firmware (ESP32)

The ESP32 firmware is responsible for real-time data acquisition, preprocessing, swing detection, and communication.

Core Functionalities:

- Sensor initialization (MPU6050)
- Continuous data acquisition
- Signal filtering and processing

5.4Data Processing & Feature Engineering

After data collection, Python is used to process and enhance the dataset.

Key Features Created:

- Power = Speed × Impact
- Efficiency = Impact / Duration
- Energy = Power × Duration

5.5 Machine Learning Layer

The system uses two trained machine learning models:

Swing Classification Model

- Input: engineered features
- Output: WEAK / MEDIUM / STRONG

Stroke Classification Model

- Input: advanced motion features
- Output: SMASH / DROP / CLEAR / DRIVE

5.6 Analytics & Decision Engine

This module performs advanced analysis on classified data.

Functions Implemented:

- Player vs Pro comparison, Gap analysis, Consistency calculation, Stability scoring, Fatigue detection, Player type classification, AI suggestions

Pseudo Code: Analytics

```
COMPUTE average speed, impact, power
COMPARE with professional dataset
CALCULATE:
   gap values
   consistency (std deviation)
   fatigue (early vs late swings)
GENERATE suggestions based on gaps
```

- o results
- o UI displays final insights

CHAPTER 6

DATA ACQUISITION AND FEATURE ENGINEERING

6.1 Introduction

Data acquisition and feature engineering play a critical role in the performance of the Smart Badminton Analytics System. The quality of data and the relevance of extracted features directly influence the accuracy of classification and analytics. This chapter describes the process of collecting motion data, structuring datasets, and transforming raw sensor readings into meaningful features for machine learning and analysis.

6.2 Data Acquisition

The system collects real-time motion data using the MPU6050 sensor integrated with the ESP32 microcontroller.

Captured Parameters:

- Accelerometer data: Ax, Ay, Az
- Gyroscope data: Gx, Gy, Gz
- Timestamp

These values are continuously read and processed to detect valid badminton swings.

6.3 Swing-Based Data Collection

Instead of storing continuous raw data, the system captures event-based swing data, which improves efficiency and relevance.

Each swing is detected using threshold-based logic and the following parameters are recorded:

- Peak speed
- Maximum impact
- Swing duration
- Processed sensor values

6.4 Dataset Structure

The collected data is stored in CSV format for further processing.

Sample Dataset Format:

timestamp, ax, ay, az, gx, gy, gz, speed, impact, duration

Each row represents a single detected swing.

6.5 Calibration Dataset and Thresholding

A calibration phase is used to establish baseline performance metrics.

Process:

- First set of swings (~30 samples) is collected
- Features are computed for calibration data
- Statistical thresholds are calculated using quantiles:
    - Weak threshold = 25th percentile
    - Strong threshold = 75th percentile

Purpose:

- Adapt system to individual player performance
- Improve classification accuracy

6.6 Feature Engineering

Feature engineering transforms raw data into meaningful parameters that improve model performance.

6.6.1 Primary Features

These are directly derived from sensor data:

- Speed → derived from angular velocity
- Impact → derived from acceleration magnitude
- Duration → time difference of swing

6.6.2 Derived Features

These features provide deeper insights:

- Power
  = Speed × Impact
- Efficiency
  = Impact / Duration
- Energy
  = Power × Duration

Pseudo Code: Feature Calculation

```
FOR each swing:
   power = speed * impact
   efficiency = impact / duration
   energy = power * duration
```

Code Concept Snippet

```
df["power"] = df["speed"] * df["impact"]
df["efficiency"] = df["impact"] / (df["duration"] + 1e-6)
df["energy"] = df["power"] * df["duration"]
```

6.7 Stroke Feature Engineering

Additional features are created specifically for stroke classification:

- Acceleration magnitude, Gyroscope magnitude, Peak acceleration, Peak angular velocity

These features help distinguish between different stroke types such as smash, drop, and clear.

6.8 Data Preprocessing

Before applying machine learning models, the dataset is cleaned and prepared.

Steps Involved:

- Removal of missing values, Selection of relevant features, Normalization (if required), Feature alignment for ML models

6.9 Importance of Feature Engineering

- Trained models are stored in .pkl format using joblib for fast deployment.
- A Swing Classification Model is used to predict stroke strength (WEAK, MEDIUM, STRONG).
- A Stroke Classification Model identifies shot types like SMASH, DROP, CLEAR, and DRIVE.
- Models are trained using labeled badminton sensor datasets.
- Input features include speed, impact, power, efficiency, acceleration, and gyroscope data.
- Algorithms used include supervised learning techniques for classification tasks.
- Models are optimized for real-time inference on embedded systems.

CHAPTER 7

MACHINE LEARNING AND HYBRID MODEL DESIGN

7.1 Introduction

Machine learning plays a crucial role in transforming raw motion data into meaningful performance insights. In this project, machine learning models are used to classify badminton swings and identify stroke types based on extracted features.

In addition to machine learning, a rule-based approach is incorporated to enhance reliability. This combination forms a hybrid intelligence system, improving both accuracy and interpretability.

7.2 Overview of Machine Learning Approach

The system uses supervised learning techniques where models are trained on labeled datasets of badminton swings.

Key Tasks:

- Swing strength classification
- Stroke type classification

Input to Models:

- Engineered features such as speed, impact, power, and energy

Output:

- Swing category (WEAK, MEDIUM, STRONG)
- Stroke type (SMASH, DROP, CLEAR, DRIVE)

7.3 Swing Classification Model

This model classifies the intensity of each swing.

Classes:

- WEAK, MEDIUM, STRONG

Features Used:

- Speed, Impact, Power, Efficiency

Working:

The trained model analyzes input features and predicts the swing category based on learned patterns.

7.4 Stroke Classification Model

This model identifies the type of badminton stroke.

Classes:

- SMASH
- DROP
- CLEAR
- DRIVE

Features Used:

- Acceleration magnitude
- Gyroscope magnitude
- Peak acceleration
- Energy

Working:

The model distinguishes stroke types based on motion patterns and feature variations.

7.5 Model Training Process

The models are trained using labeled datasets collected during experimentation.

Steps Involved:

1. Data collection and labeling
2. Feature engineering
3. Splitting dataset into training and testing sets
4. Model training using appropriate algorithm
5. Performance evaluation

Model Storage:

- Trained models are saved using serialization (.pkl files)
- Loaded during prediction phase

7.6 Hybrid Model Design

A hybrid approach is implemented by combining machine learning predictions with rule-based classification.

Why Hybrid?

- Improves robustness
- Handles edge cases
- Provides interpretable logic

7.7 Model Performance

The system achieves high classification accuracy based on experimental results.

Observed Performance:

- Accuracy ≈ 98% for swing classification
- Reliable stroke classification performance

Factors Contributing to Accuracy:

- Effective feature engineering
- Clean dataset
- Hybrid decision logic

7.8 Advantages of Machine Learning Approach

- Enables automatic pattern recognition
- Reduces dependency on manual thresholds
- Adapts to different player styles
- Supports scalable and extensible design

CHAPTER 8

SYSTEM IMPLEMENTATION AND ALGORITHM DESIGN

8.1 Introduction

This chapter describes the practical implementation of the Smart Badminton Performance Analytics System. It covers the integration of hardware, software, machine learning models, and user interfaces. Additionally, the algorithms used for swing detection, feature extraction, and classification are explained.

8.2 Implementation Overview

The system is implemented in three main modules:

- Embedded Module (ESP32)
- Data Processing & Machine Learning Module (Python)
- User Interface Module (Web Dashboard & Streamlit)

Each module works independently and communicates through structured data.

8.3 Embedded System Implementation (ESP32)

The ESP32 firmware is responsible for real-time data acquisition, swing detection, and communication.

Key Tasks Implemented:

- Sensor initialization using I2C
- Continuous data reading from MPU6050
- Gravity filtering and signal processing
- Swing detection using threshold logic
- Data logging in CSV format
- Hosting a web server for real-time monitoring

Algorithm 1: Swing Detection Algorithm

The system detects a valid swing based on acceleration thresholds and time duration.

```
INPUT: Accelerometer and gyroscope data
WHILE system is running:
   read sensor data
   compute total acceleration
```

```
    IF totalAcceleration > START_THRESHOLD:
        start swing
        record start time
    WHILE swing is active:
        track peak speed and impact
    IF totalAcceleration < END_THRESHOLD AND duration >
MIN_DURATION:
        end swing
        store swing data
```

8.4 Feature Extraction Implementation

After detecting a swing, key parameters are computed.

Algorithm 2: Feature Extraction

```
FOR each detected swing:
    speed = function(angular velocity)
    impact = function(acceleration)
    duration = end_time - start_time

    power = speed * impact
    efficiency = impact / duration
```

Code Concept

```
power = speed * impact
efficiency = impact / (duration + 1e-6)
```

8.5 Machine Learning Implementation

The system uses trained models stored as .pkl files.

Algorithm 3: Swing Classification

```
LOAD swing_model
FOR each swing:
    input_features = [speed, impact, power, efficiency]
    prediction = swing_model.predict(input_features)
```

Algorithm 4: Stroke Classification

```
LOAD stroke_model
FOR each swing:
    input_features = [acc_mag, gyro_mag, peak_acc, energy]
    stroke_type = stroke_model.predict(input_features)
```

8.6 Hybrid Classification Implementation

Algorithm 5: Rule-Based Classification

```
IF power < weak_threshold:
   label = WEAK
ELSE IF power < strong_threshold:
   label = MEDIUM
ELSE:
   label = STRONG
```

Final Decision Strategy

```
IF ML prediction is reliable:
   use ML result
ELSE:
   use rule-based result
```

8.7 WORKING OF THE IMPLEMENTED ALGORITHMS

Swing Detection Working

The system continuously reads real-time accelerometer and gyroscope data from the IMU sensor. The swing detection algorithm identifies the start of a valid badminton stroke when the total acceleration exceeds a predefined threshold value. Once a swing is initiated, the system records the start time and continuously tracks motion parameters such as peak speed and impact force.

Feature Extraction Working

Once a valid swing is detected, the system computes meaningful performance features from raw sensor data. Speed is derived from angular velocity, while impact is calculated from acceleration magnitude. The duration is obtained using the difference between start and end timestamps.

Machine Learning Classification Working

The extracted features are then passed to trained machine learning models stored in .pkl format. The swing classification model predicts the strength of the shot as WEAK, MEDIUM, or STRONG based on performance metrics like speed, impact, power, and efficiency.

In parallel, the stroke classification model identifies the type of shot such as SMASH, DROP, CLEAR, or DRIVE using features like acceleration magnitude, gyroscope magnitude, peak acceleration, and energy. This enables detailed analysis of player behavior and technique.

CHAPTER 9

RESULTS AND PERFORMANCE ANALYSIS

9.1 Introduction

This chapter presents the experimental results obtained from the Smart Badminton Performance Analytics System. It evaluates system performance in terms of accuracy, real-time capability, and analytical insights generated from player data.

9.2 Experimental Setup

- Hardware: ESP32 + MPU6050 mounted on racket
- Modes Used: Calibration and Game mode
- Data Collected: Multiple swing sessions (event-based)
- Processing: Python pipeline with trained ML models
- Visualization: ESP32 Web Dashboard and Streamlit UI

9.3 Dataset Summary

- Total swings recorded: 102
- Features per swing: speed, impact, duration, power, efficiency
- Dataset type: event-based CSV (one row per swing)
- Calibration samples: 30 swings

Stroke Classification Performance

- Successfully classified strokes into:
  - SMASH, DROP, CLEAR, DRIVE

Observation:

- Distinct motion patterns enabled effective classification
- Feature engineering significantly improved accuracy

9.4 Real-Time System Performance

- Data acquisition delay: very low (~milliseconds)
- Dashboard refresh rate: ~200 ms
- Smooth real-time monitoring achieved

Observation:

- No significant lag during swing detection
- Stable performance during continuous sessions

9.5 Visualization Results

ESP32 Dashboard

- Displays real-time: Speed, Impact, Swing count

Streamlit Dashboard

- Displays: Performance metrics, Stroke distribution, AI suggestions, Graphical analysis

9.6 Comprehensive session report

SESSION OVERVIEW

Metric	Value
Total Swings	102
Weak Swings	19
Medium Swings	63
Strong Swings	20
Player Level	Needs Major Improvement
Player Type	Attacker

Fig 9.1: Session overview

PERFORMANCE METRICS

Metric	Your Average	Pro Average	Gap
Speed	47.88	53.51	-5.63
Impact	34.72	47.35	-12.63
Power	1700.34	2532.58	-832.24
Efficiency	77.73	69.01	8.71

Fig 9.2: Performance metrics

FATIGUE & CONSISTENCY

Metric	Value
Fatigue Status	Fresh performance throughout session
Fatigue Score	-11.04%
Speed Drop	-9.65%
Impact Drop	-7.48%
Power Drop	-16.00%
Speed CV	0.156
Impact CV	0.169
Power CV	0.261
Consistency Score	80.47/100

Fig: 9.3: Fatigue and consistence report

9.10 System Strengths

- High classification accuracy
- Real-time processing capability
- Hybrid ML + rule-based reliability
- Advanced analytics and insights
- User-friendly visualization

CHAPTER 10

FUTURE SCOPE

10.1 Introduction

The current Smart Badminton Performance Analytics System has been developed as a functional prototype using ESP32 and MPU6050 for real-time motion tracking and machine learning-based performance analysis. While the system demonstrates strong accuracy and usability, it is primarily a lab-scale implementation.

10.2 Transition from Prototype to Product

The productization process focuses on reducing size and weight, improving power efficiency, and upgrading hardware components for wearable deployment inside a badminton racket.

10.3 Proposed Hardware Upgrades

Microcontroller Upgrade

- From ESP32 Dev Board, To: Ultra-low power compact boards such as:
    - Seeed Studio XIAO nRF52840, ESP32-C3

Sensor Upgrade

- From MPU6050, To: ICM42688 (high-performance IMU sensor) for Better accuracy, lower noise, improved stability

Power System Upgrade

- From standard battery setup
- To: 200mAh ultra-compact Li-Po battery

10.4 COMPARISON: PROTOTYPE vs FUTURE PRODUCT

Parameter	Current Prototype (ESP32 + MPU6050)	Future Product (XIAO nRF52840 / ESP32-C3 + ICM42688)
Microcontroller Weight	~25–30 g	~3–10 g
Sensor Weight	~4–5 g (MPU6050 module)	~2–3 g (ICM42688 module)

Parameter	Current Prototype (ESP32 + MPU6050)	Future Product (XIAO nRF52840 / ESP32-C3 + ICM42688)
Battery Weight	~11–20 g	~4–6 g (200mAh Li-Po)
Total System Weight	~45–80 g	**~10–20 g**
Power Consumption	High (WiFi active mode)	**Very Low** (BLE / optimized sleep modes)
Communication	WiFi-based streaming	BLE (low power, mobile friendly)
Accuracy	Good (prototype level)	**Higher** (low-noise sensor + better IMU)

10.5Key Improvements in Product Version

- **Weight Reduction:** Nearly **80–90% reduction in total system weight**
- **Power Efficiency:** BLE communication instead of continuous WiFi, Deep sleep modes for microcontroller, Ultra-low power IMU (ICM42688)
- **Sensor Accuracy:** Reduced drift and noise, Better motion tracking resolution, Improved stroke classification reliability

10.6 Future Product Vision

The upgraded system can evolve into a:

- Smart badminton racket with embedded AI
- Wearable sports performance tracker
- Mobile-connected training ecosystem

PROJECT SUMMARY

The Smart Badminton Performance Analytics System is an intelligent solution designed to analyze player performance using embedded systems and machine learning techniques. The system integrates an ESP32 microcontroller with an MPU6050 inertial measurement unit (IMU) to capture real-time motion data such as acceleration and angular velocity during badminton swings. A threshold-based algorithm is used to detect valid swings, ensuring accurate event-based data collection.

Key parameters including speed, impact, and duration are extracted and further processed through feature engineering to derive advanced metrics such as power, efficiency, and energy. These features are used as inputs for machine learning models that classify swings into WEAK, MEDIUM, and STRONG categories, and identify stroke types such as SMASH, DROP, CLEAR, and DRIVE. A hybrid approach combining machine learning with rule-based classification improves overall system reliability and interpretability.

The processed data is analyzed to generate meaningful insights including player performance level, consistency score, fatigue detection, and comparison with professional benchmarks. The system provides real-time feedback through a web-based dashboard and a Streamlit interface, enabling effective visualization of performance metrics.

The project demonstrates high accuracy, efficient real-time processing, and a scalable architecture. Furthermore, it outlines future enhancements for product development, including the use of low-power microcontrollers, advanced sensors, and compact battery systems to enable a lightweight, wearable smart sports device for practical deployment.

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